L-MOMENT BASED FLOOD FREQUENCY MODELLING OF ANNUAL MAXIMUM AND DAILY RAINFALL DATA IN TERENGGANU

ABSTRACT

Introduction: Flood frequency analysis is essential for water resource management, hydraulic infrastructure design, and disaster risk reduction, particularly in tropical regions prone to recurrent monsoon floods. Conventional flood frequency analysis relies primarily on the Annual Maximum Series (AMS) framework paired with standard probability distributions formulated for temperate regions. In tropical monsoon climates such as Terengganu, Malaysia, this conventional approach causes information loss by discarding other extreme rainfall events occurring within the same year. Furthermore, standard distributions fail to model such intense tropical storms. Consequently, this study applies the robust L-moments method to address these limitations by comparing the traditional AMS approach with Complete Time-series Analysis (CTA) using daily rainfall data across 20 hydrological stations in Terengganu. **Methodology:** The research procedure involved four phases. First, daily rainfall datasets from 20 gauging stations were compiled and processed into both AMS and full daily time-series records. Second, sample L-moments (λ_1, λ_2) and L-moment ratios (L-skewness τ_3 and L-kurtosis τ_4) were computed to characterize regional data properties. Third, parameters for nine candidate probability distributions (Gumbel, Normal, Exponential, Generalized Extreme Value, Generalized Logistic, Generalized Normal, Generalized Pareto, Pearson Type III, and 4-Parameter Kappa) were estimated using L-moments. Distribution fit was evaluated using the Mean Absolute Deviation Index (MADI) and Mean Squared Deviation Index (MSDI) against Gringorten plotting positions. Finally, design rainfall quantiles were estimated for return periods between 2 and 100 years, and an overestimation factor was formulated to quantify the recurrence bias between AMS and daily rainfall series. **Results:** The empirical analysis demonstrated that the 4-Parameter Kappa distribution provided the optimal fit across both datasets, yielding the best fit for 40% of AMS stations and 100% of daily stations, which confirms its suitability for skewed tropical rainfall. Daily rainfall records exhibited higher L-skewness ($\tau_3 \approx 0.47–0.55$) relative to the AMS series, confirming positive skewness that standard models underestimate. The traditional AMS approach exhibited systematic overestimation of return periods, yielding an average overestimation factor of 8.50 at the 99th percentile relative to daily rainfall data. This discrepancy demonstrates that an extreme rainfall event classified under AMS as a 1-year recurrence event occurs approximately every 43 days when evaluated using complete daily observations. **Conclusion:** The findings demonstrate that traditional AMS approaches misjudge the recurrence frequency of extreme rainfall in tropical environments due to annual data reduction. Flood frequency analysis in tropical monsoon regions improves when utilizing complete daily rainfall series coupled with the 4-Parameter Kappa distribution. These results provide practical guidance for

hydraulic infrastructure design and flood hazard mitigation for tropical climate countries.

PEMODELAN KEKERAPAN BANJIR BERASASKAN MOMEN-L BAGI DATA HUJAN MAKSIMUM TAHUNAN DAN HARIAN DI TERENGGANU

ABSTRAK

Pengenalan: Analisis kekerapan banjir merupakan asas penting dalam pengurusan sumber air, reka bentuk infrastruktur hidraulik, dan pengurangan risiko bencana, terutamanya di kawasan tropika yang terdedah kepada banjir monsun berulang. Analisis kekerapan banjir konvensional bergantung kepada kerangka Siri Maksimum Tahunan (AMS) yang dipadankan dengan taburan kebarangkalian piawai yang asalnya dirumuskan untuk kawasan beriklim sederhana. Di kawasan beriklim monsun tropika seperti Terengganu, Malaysia, pendekatan konvensional ini menyebabkan kehilangan data kerana mengabaikan kejadian hujan melampau lain yang berlaku dalam tahun yang sama. Selain itu, taburan piawai gagal memodelkan kejadian ribut tropika yang melampau ini. Oleh itu, kajian ini mengaplikasikan kaedah momen-L yang teguh bagi menangani batasan tersebut dengan membandingkan pendekatan tradisional AMS dengan Analisis Siri Masa Lengkap (CTA) menggunakan data hujan harian merentasi 20 stesen hidrologi di Terengganu. **Metodologi:** Kajian ini dijalankan melalui empat fasa. Pertama, data hujan daripada 20 stesen tolok dikumpulkan dan diproses kepada rekod AMS dan siri masa harian yang lengkap. Kedua, momen-L sampel (λ_1, λ_2) dan nisbah momen-L (kepencongan-L τ_3 dan kurtosis-L τ_4) dikira bagi mencirikan sifat data serantau. Ketiga, parameter bagi sembilan calon taburan kebarangkalian (Gumbel, Normal, Eksponen, Nilai Ekstrem Teritlak, Logistik Teritlak, Normal Teritlak, Pareto Teritlak, Pearson Jenis III, dan Kappa 4-Parameter) dianggar menggunakan kaedah momen-L. Penyuaian taburan dinilai menggunakan Indeks Sisihan Mutlak Min (MADI) dan Indeks Sisihan Kuasa Dua Min (MSDI) bersandarkan kedudukan pemplotan Gringorten. Akhir sekali, kuantil hujan reka bentuk dianggar bagi kala ulangan antara 2 hingga 100 tahun, dan faktor lebihan anggaran dirumuskan bagi mengkuantifikasi pincang penentuan keberulangan antara siri AMS dan siri hujan harian. **Keputusan:** Analisis empirikal membuktikan bahawa taburan Kappa 4-Parameter memberikan penyuaian paling optimum bagi kedua-dua set data, iaitu sebagai taburan terbaik untuk 40% stesen AMS dan 100% stesen harian, sekali gus mengesahkan kesesuaian taburan tersebut bagi corak hujan tropika yang terpencong. Rekod hujan harian menunjukkan kepencongan-L yang lebih tinggi ($\tau_3 \approx 0.47–0.55$) berbanding siri AMS, mengesahkan ciri terpencong ke kanan di mana model piawai lazimnya memberikan anggaran terkurang. Pendekatan tradisional AMS menunjukkan lebihan anggaran kala ulangan yang sistematik, dengan purata faktor lebihan anggaran sebanyak 8.50 pada persentil ke-99 berbanding data hujan harian. Percanggahan ini membuktikan bahawa sesuatu kejadian hujan melampau yang diklasifikasikan di bawah AMS sebagai kejadian berulang sekali setahun sebenarnya berlaku kira-kira sekali setiap 43 hari apabila dinilai menggunakan cerapan harian yang lengkap. **Kesimpulan:** Penemuan ini membuktikan bahawa pendekatan tradisional AMS terkurang anggar

terhadap kekerapan sebenar kejadian hujan melampau di kawasan tropika akibat pengabaian data harian. Analisis kekerapan banjir di kawasan monsun tropika bertambah baik melalui penggunaan siri data hujan harian yang lengkap bersama taburan Kappa 4-Parameter. Penemuan ini menyediakan panduan praktikal untuk reka bentuk infrastruktur hidraulik dan mitigasi bahaya banjir untuk negara beriklim tropika.